

Phase 1-Configure GoldenGate Replication from Oracle Range to Oracle Hash

- Setup and configure replication from Range RDS to Hash RDS
 - Setup Environment
 - Configure GG Replication from Range Partitioning to Hash Partitioning
 - Create New GG Hub to setup replication for Hash RDS
 - Create and start Pump to send trails to HASH GG hub
 - Restore latest snapshot of Range RDS via console.
 - Create New HASH RDS
 - Initial Sync
 - Export data from PITHASH1 db
 - Import data to HASH RDS db
 - Final step for Goldengate replication from Range to Hash RDS
- Setup GoldenGate replication from Hash RDS (B) to Range RDS (C)
 - Final step for goldengate replication from Hash db(B) to Range db(C)

- Setup and configure replication from Range RDS to Hash RDS
- Setup Environment
- Configure GG Replication from Range Partitioning to Hash Partitioning
- Create New GG Hub to setup replication for Hash RDS

```
./create_gg_hub_ha_new.sh psp prod pspphp01 us-west-2 vpc-2 no m4.10xlarge 1000
```

- Create and start Pump to send trails to HASH GG hub
- Login to current production GG hub and run below commands

```
GGSCI (ggpspuwp01.sbg-ppsp-prod.a.intuit.com) 2> obey ./diroby/create_ppsphp01.oby
```

```
GGSCI (ggpspuwp01.sbg-ppsp-prod.a.intuit.com) 3> add extract ppsphp01, EXTTRAILSOURCE ./dirdat/pspuwp01/tr
```

```
EXTRACT added.
```

```
GGSCI (ggpspuwp01.sbg-ppsp-prod.a.intuit.com) 4> add rmttrail ./dirdat/pspuwp01/tr, extract ppsphp01, MEGABYTES 500
```

```
RMTTRAIL added.
```

```
GGSCI (ggpspuwp01.sbg-ppsp-prod.a.intuit.com) 5> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPWP01	00:00:02	00:00:00
EXTRACT	RUNNING	EUIPWP01	00:00:03	00:00:03
EXTRACT	ABENDED	PPSPEP01	00:00:00	00:04:40
EXTRACT	RUNNING	PPSPEP05	00:00:03	00:00:08
EXTRACT	STOPPED	PPSPHP01	00:00:00	00:00:05
EXTRACT	RUNNING	PUIPWP01	00:00:03	00:00:09

- Restore latest snapshot of Range RDS via console.

```
--get SCN
SQL> sho parameter db_name
```

NAME	TYPE	VALUE
db_name	string	PITHASH1

```
SQL> select name from v$database;
```

NAME
PITHASH1

```
SQL> select incarnation#, resetlogs_change#, resetlogs_time, prior_resetlogs_change#, prior_resetlogs_time,
status, resetlogs_id, prior_incarnation# from v$database_incarnation;
```

PRIOR_RES STATUS	INCARNATION#	RESETLOGS_CHANGE#	RESETLOGS_ID	RESETLOGS PRIOR_INCARNATION#	PRIOR_RESETLOGS_CHANGE#
20 PARENT	1	8388795358333	1048411759	14-AUG-20	8388795357611 14-AUG-
20 PARENT	2	8397003383021	1133501356	07-APR-23	8388795358333 14-AUG-
23 CURRENT	3	8397003385952	1133501971	07-APR-23	8397003383021 07-APR-

```
SQL> select current_scn from v$database;
```

CURRENT_SCN
8397003404258

- Increase RDS storage for PITHASH1 to have ~14TB free for export backup & Take full database export in above restored RDS

• Create New HASH RDS

```
./create_rds.sh prodx clusterdb psphp02 us-west-2 vpc-2

--Change Master user password
aws --profile sbg-psp-prod --region us-west-2 rds modify-db-instance --db-instance-identifier psphp02 --master-user-password xxxxxxxx
```

- Initial Sync
- **Export data from PITHASH1 db**
- create directories in PITHASH1 db & Hash rds

```
EXEC rdsadmin.rdsadmin_util.create_directory(p_directory_name => 'DATA_PUMP_DIR1');
EXEC rdsadmin.rdsadmin_util.create_directory(p_directory_name => 'DATA_PUMP_DIR2');
EXEC rdsadmin.rdsadmin_util.create_directory(p_directory_name => 'DATA_PUMP_DIR3');
select * from DBA_DIRECTORIES where DIRECTORY_NAME like 'DATA_PUMP_DIR%';
```

- create database link in PITHASH1 db

```
drop database link to_target;
create database link to_target connect to intuadmin identified by "xxxxxx" using '(DESCRIPTION=(ADDRESS=
(PROTOCOL=TCP)(HOST= psphp02.sbg-psp-prod.a.intuit.com)(PORT=1521))(CONNECT_DATA=(SID=psphp02)))';
select INSTANCE_NAME, HOST_NAME from v$instance@to_target;
```

- export metadata

```
nohup expdp userid=intuadmin/xxxxx@pithash1 directory=DATA_PUMP_DIR dumpfile=exp_pspadm_metadata.dmp
SCHEMAS=PSPADM content=METADATA_ONLY logfile=DATA_PUMP_DIR:exp_pspadm_metadata.log 2>&1 &
```

- export data

```
nohup expdp userid=intuadmin/"xxxx"@pithash1 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,DATA_PUMP_DIR1:exp_data1_%U.
dmp,DATA_PUMP_DIR2:exp_data2_%U.dmp,DATA_PUMP_DIR3:exp_data3_%U.dmp parallel=48 schemas=PSPADM filesize=150G
content=data_only logfile=DATA_PUMP_DIR:exp_data.log 2>&1 &
```

```
--export dump size : 13.893 TB (~14 TB) (Time Taken: ~11 hrs)
```

- Copy dump files to target Hash db

```
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p1.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p2.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p3.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p4.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p5.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p6.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p7.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p8.sql &
```

```
--Overall time taken ~8 hrs
```

• Import data to HASH RDS db

- Change DB parameter group to have new undo retention as 900 (15 mins)
- Recreate redo logs

```
./run_on_target.sh add_redos.sql
```

```
--ensure no errors
grep ORA- add_redos.lst
```

- create tablespaces on Hash RDS

```
nohup ./run_on_target.sh create_tablespaces.sql &
```

```
--ensure no errors
grep ORA- create_tablespaces.lst
```

```
--Temporarily change TEMP and UNDO tablespace size
alter tablespace temp resize 5T;
alter tablespace UNDO_T1 resize 4T;
```

- Create schema owners

```
./run_on_target.sh create_schema_owners.sql
```

- Create tables with PK

```
./run_sql.sh pspadm_hash_Partitioned_tables_ddl.sql
./run_sql.sh tables.sql
./run_sql.sh create_pk.sql
```

- import data

```

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,DATA_PUMP_DIR1:exp_data1_%
U.dmp,DATA_PUMP_DIR2:exp_data2_%U.dmp,DATA_PUMP_DIR3:exp_data3_%U.dmp parallel=16 schemas=PSPADM exclude=TABLE:
\IN \(\PSP_FINANCIAL_TRANS_STATE\,'PSP_PSTUB_PAY_ITEM\','PSP_ENTRY_DETAIL_RECORD\','PSP_TAX\',
'PSP_COMPANY_EVENT_EMAIL_PARAM\','PSP_QBDT_PAYLINE_INFO\','PSP_COMPANY_EVENT_DETAIL\',
'PSP_ENTITY_UPDATE_HIST\','PSP_SOURCE_SYSTEM_TRANSMISSION\','T_PPPI_OLD\','T_PCEEP_OLD\','T_PDATL_OLD\',
'T_PPU_OLD\','T_PPPI_OLD\','T_PPU\')\)" content=data_only EXCLUDE=STATISTICS
TRANSFORM=DISABLE_ARCHIVE_LOGGING:Y logfile=DATA_PUMP_DIR:imp_data.log 2>&1 &
--Time taken ~15 hrs

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,DATA_PUMP_DIR1:exp_data1_%
U.dmp,DATA_PUMP_DIR2:exp_data2_%U.dmp,DATA_PUMP_DIR3:exp_data3_%U.dmp parallel=16 TABLES=PSPADM.
PSP_FINANCIAL_TRANS_STATE,PSPADM.PSP_PSTUB_PAY_ITEM content=data_only EXCLUDE=STATISTICS
TRANSFORM=DISABLE_ARCHIVE_LOGGING:Y logfile=DATA_PUMP_DIR:imp_psp_tables1.log 2>&1 &
--Time taken 16 hrs 30 mins

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,DATA_PUMP_DIR1:exp_data1_%
U.dmp,DATA_PUMP_DIR2:exp_data2_%U.dmp,DATA_PUMP_DIR3:exp_data3_%U.dmp parallel=16 TABLES=PSPADM.
PSP_ENTRY_DETAIL_RECORD,PSPADM.PSP_TAX content=data_only EXCLUDE=STATISTICS TRANSFORM=DISABLE_ARCHIVE_LOGGING:Y
logfile=DATA_PUMP_DIR:imp_psp_tables2.log 2>&1 &
--Time taken ~11 hrs 45 mins

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,DATA_PUMP_DIR1:exp_data1_%
U.dmp,DATA_PUMP_DIR2:exp_data2_%U.dmp,DATA_PUMP_DIR3:exp_data3_%U.dmp parallel=16 TABLES=PSPADM.
PSP_COMPANY_EVENT_EMAIL_PARAM,PSPADM.PSP_QBDT_PAYLINE_INFO,PSPADM.PSP_COMPANY_EVENT_DETAIL content=data_only
EXCLUDE=STATISTICS TRANSFORM=DISABLE_ARCHIVE_LOGGING:Y logfile=DATA_PUMP_DIR:imp_psp_tables3.log 2>&1 &
--Time taken ~5 hrs 20 mins

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,DATA_PUMP_DIR1:exp_data1_%
U.dmp,DATA_PUMP_DIR2:exp_data2_%U.dmp,DATA_PUMP_DIR3:exp_data3_%U.dmp parallel=16 TABLES=PSPADM.
PSP_FINANCIAL_TRANSACTION,PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION content=data_only EXCLUDE=STATISTICS
TRANSFORM=DISABLE_ARCHIVE_LOGGING:Y logfile=DATA_PUMP_DIR:imp_psp_tables4.log 2>&1 &
--Time taken ~4 hrs 30 mins

--Ensure No Errors in all import logs

```

- run row count sanity (in case of import failures seen in import log)
- collect stats

```

nohup ./run_sql.sh collect_stats.sql &
--Time taken ~6 hrs 30 mins

```

- cleanup dump files

```

nohup ./run_sql.sh cleanup_exp_dump_files.sql &

```

- Create other Indexes

```

./run_sql.sh create_sec_fk_indexes_1.sql
./run_sql.sh create_sec_fk_indexes_2.sql
./run_sql.sh create_sec_fk_indexes_3.sql
./run_sql.sh create_sec_fk_indexes_4.sql

```

```

--Time Taken ~27 hrs

```

- (Optional) Below is optional in case there is any error while index creation

```

CREATE INDEX PSPADM.PSP_PSTUB_PAY_ITEM_FK1 ON PSPADM.PSP_PSTUB_PAY_ITEM (COMPANY_FK,PAYSTUB_FK) local online
parallel (degree 4) TABLESPACE PSP_IDX01;

--got below error for few indexes, so created indexes offline as below(without using online keyword)
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--offline rebuild
CREATE INDEX PSPADM.PSP_AGENCY_AGENCYIDENC_I1 ON PSPADM.PSP_AGENCY (AGENCY_ID_ENC) parallel (degree 4)
TABLESPACE PSP_IDX02;
CREATE INDEX PSPADM.PSP_BANK_ACCOUNT_ENC_I1 ON PSPADM.PSP_BANK_ACCOUNT (ACCOUNT_NUMBER_ENC) parallel (degree 4)
TABLESPACE PSP_IDX01;
CREATE INDEX PSPADM.PSP_COMPANY_FEDTAXIDENC_I1 ON PSPADM.PSP_COMPANY (FED_TAX_ID_ENC) parallel (degree 4)
TABLESPACE PSP_IDX02;

CREATE INDEX PSPADM.PSP_EFTPS_PAYMENT_DETAIL_I3 ON PSPADM.PSP_EFTPS_PAYMENT_DETAIL (EFT_TRANSACTION_ID)
parallel (degree 4) TABLESPACE PSP_IDX02;
CREATE INDEX PSPADM.PSP_EFTPS_PAYMENT_DETAIL_I4 ON PSPADM.PSP_EFTPS_PAYMENT_DETAIL (TRANSACTION_ID,
AGENCY_PAYMENT_ID) parallel (degree 4) TABLESPACE PSP_IDX02;
CREATE INDEX PSPADM.PSP_EMPLOYEE_TAXIDENC_I1 ON PSPADM.PSP_EMPLOYEE (TAX_ID_ENC) parallel (degree 4)
TABLESPACE PSP_IDX01;
CREATE INDEX PSPADM.PSP_ENMT_FEDTAXIDENC_I1 ON PSPADM.PSP_ENTITLEMENT_UNIT (FED_TAX_ID_ENC) parallel (degree
4) TABLESPACE PSP_IDX01;

CREATE INDEX PSPADM.PSP_FRAUD_ACCOUNT_ENC_I1 ON PSPADM.PSP_FRAUD_BANK_ACCOUNT (ACCOUNT_NUMBER_ENC) parallel
(degree 4) TABLESPACE PSP_IDX01;

```

- disable index parallel degree

```
./run_sql.sh disable_index_parallel.sql &
```

- create FK constraints

```
./run_sql.sh ref_constraints.sql
```

NOTE:

No FK between PSP_FINTXN_ONHOLDREASON_ASSOC & PSP_FINANCIAL_TRANSACTION
No FK between PSP_PAYCHECK_USAGE_HIST & PSP_PAYCHECK_USAG

- Import other objects

```

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_pspadm_metadata.dmp schemas=PSPADM
EXCLUDE=TABLE,INDEX,REF_CONSTRAINT,STATISTICS content=METADATA_ONLY TRANSFORM=DISABLE_ARCHIVE_LOGGING:Y
logfile=DATA_PUMP_DIR:imp_pspadm_metadata.log 2>&1 &

```

```

impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_pspadm_metadata.dmp include=trigger
logfile=DATA_PUMP_DIR:imp_pspadm_triggers.log

```

- create check and not null constraints via generated script from source(production)

```
./run_sql.sh create_notnull_cc.sql
```

--Ensure no errors

- create users

```
./run_sql.sh create_users.sql
```

--below should not return any PSPADM objects errors

```
grep -B 3 ORA-00942 CREATE_USERS.lst|egrep -v '"PSPBOADMIN"."|"APPBUGFIX"."|"MCHOUBEY"."|"SYS"."|"DIYMIGADM"."|"  
PSPIDPSBACKUP"."|"SKUMAR71"."|"PSPADM"."PSP_SOURCE_SYSTEM_TRANSMISSION"|PSPLOG"."|"PSPFLUXADM"."|"PSPTEMP"."|'
```

- lock app users

```
alter user pspapp account lock;  
alter user pspadm account lock;
```

- check for invalid objects.

```
select owner, object_type, object_name from dba_objects where status <> 'VALID';
```

- Run Sanity check on source and target HASH rds db

```
@sanity_chk.sql
```

- Final step for Goldengate replication from Range to Hash RDS
- login to HASH RDS
- Setup supplemental log

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;  
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');  
exec rdsadmin.rdsadmin_util.force_logging();  
exec rdsadmin.rdsadmin_util.switch_logfile;  
exec rdsadmin.rdsadmin_util.set_configuration('archive log retention hours',72);  
commit;
```

- Grant permission to GGT and GGS

```
grant create session to GGT;  
grant DBA to GGT;  
exec rdsadmin.rdsadmin_dbms_goldengate_auth.grant_admin_privilege(grantee=>'GGT',grant_select_privileges=>true,  
do_grants=>TRUE);  
grant exempt access policy to GGT;
```

```
GRANT SELECT ANY DICTIONARY TO GGS;  
host sed '8,25d' $GG_HOME/sequence.sql > $GG_HOME/rds_sequence.sql  
@$GG_HOME/rds_sequence.sql ggs  
GRANT EXECUTE ON GGS.updateSequence to GGT;  
GRANT EXECUTE ON GGS.replicateSequence to GGT;  
drop table GGT.CHECKPOINT;  
drop table GGT.CHECKPOINT_LOX;
```

- Create credentials and checkpoint.

```
ALTER CREDENTIALSTORE ADD USER ggs@psphpp02 PASSWORD "xxxxx" ALIAS ggsource_psphpp02;  
ALTER CREDENTIALSTORE ADD USER ggt@psphpp02 PASSWORD "xxxxx" ALIAS ggtarget_psphpp02;  
obey ./diroby/dblogin_s_psphpp02.oby  
obey ./diroby/dblogin_t_psphpp02.oby  
add checkpointtable
```

- add schema trandata

```
obey ./diroby/dblogin_s.oby
add schematrandata PSPADM
```

- Setup and start oracle range to oracle hash replicat process

```
obey ./diroby/create_rpswp01.oby

GGSCI (ggpsphpp01.sbg-ppsp-prod.a.intuit.com) 2> obey ./diroby/create_rpswp01.oby

GGSCI (ggpsphpp01.sbg-ppsp-prod.a.intuit.com) 3> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpsphpp01.sbg-ppsp-prod.a.intuit.com as ggt@PSPHPP01) 4> add replicat rpswp01, INTEGRATED, EXTTRAIL .
/dirdat/pspuwp01/tr

REPLICAT (Integrated) added.

GGSCI (ggpsphpp01.sbg-ppsp-prod.a.intuit.com as ggt@PSPHPP01) 11> register replicat rpswp01 database

2023-03-31 06:29:18 INFO      OGG-02528  REPLICAT RSPWP01 successfully registered with database as inbound
server OGG$RSPWP01.
```

- Get correct SCN from PITHASH1 db and modify below. Start replication atcsn

```
start rpswp01 atcsn 8397003383021
```

Setup GoldenGate replication from Hash RDS (B) to Range RDS (C)

- [Shut down application & ensure no application connections](#)
- On Hash GG hub, create extract and pump [to Range RDS (C)]

```
obey ./diroby/create_epsphp02.oby
obey ./diroby/dblogin_s_psp02.oby
register extract epsphp02

start epsphp02
info epsphp02

obey ./diroby/create_ppsprp01.oby

start ppsprp01
info ppsprp01
```

- Verify trails are coming to Range GG hub

```
ls $GG_HOME/dirdat/psphpp02/
```

- Take Snapshot of Production (A)
- Once snapshot completed, Start applications back
- Restore snapshot as new Range RDS (C)

Final step for goldengate replication from Hash db(B) to Range db(C)

- [login to Range db \(C\)](#)

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');
exec rdsadmin.rdsadmin_util.force_logging();
exec rdsadmin.rdsadmin_util.switch_logfile;
exec rdsadmin.rdsadmin_util.set_configuration('archive log retention hours',72);
commit;
```

- [Grant permission to GGT and GGS](#)

```
grant create session to GGT;
grant DBA to GGT;
exec rdsadmin.rdsadmin_dbms_goldengate_auth.grant_admin_privilege(grantee=>'GGT',grant_select_privileges=>true,
do_grants=>TRUE);
grant exempt access policy to GGT;
GRANT SELECT ANY DICTIONARY TO GGS;
```

- [run rds_sequences.sql on Range db \(C\)](#)

```
host sed '8,25d' $GGG_HOME/sequence.sql > $GGG_HOME/rds_sequence.sql
@$GGG_HOME/rds_sequence.sql ggs
GRANT EXECUTE ON GGS.updateSequence to GGT;
GRANT EXECUTE ON GGS.replicateSequence to GGT;
drop table GGT.CHECKPOINT;
drop table GGT.CHECKPOINT_LOX;
```

- [create credentialstore on Range \(C\) gg hub](#)

```
ALTER CREDENTIALSTORE ADD USER ggs@psprpp01 PASSWORD "WGgRgK7d#(7eX" ALIAS ggsources;
ALTER CREDENTIALSTORE ADD USER ggt@psprpp01 PASSWORD "WGgRgK7d#(7eX" ALIAS ggtarget;
obey ./diroby/dblogin_s.oby
obey ./diroby/dblogin_t.oby
add checkpointtable
```

- [add schematrandata](#)

```
obey ./diroby/dblogin_s.oby
add schematrandata PSPADM
```

- On Range GG hub, create replicat

```
obey ./diroby/create_rpsphp02.oby
register replicat rpsphp02 database
```

- On Range GG Hub, start replicat

```
start rpsphp02
```

- On Hash GG hub, Run following commands to flush sequences

```
GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com) 5> dblogin useridentialias ggsources_hash_pds
Successfully logged into database.

GGSCI (ggpsppf501.sbg-ppsp-ppd.a.intuit.com as ggs@PPSPHP01) 6> flush sequence pspadm.*

2023-05-18 01:09:31 INFO OGG-15311 Successfully flushed 76 sequence(s) pspadm.*.
```

- Setup GG monitoring on Range GG hub

- verify sequence bi-directional replication

1. On Range (current live database)

```
SQL> create sequence pspadm.seq_test;
```

Sequence created.

2. SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER = 'PSPADM' and SEQUENCE_NAME = 'SEQ_TEST';

SEQUENCE_OWNER	SEQUENCE_NAME	LAST_NUMBER
PSPADM	SEQ_TEST	1

3. Test the sequence

a) Run it on Range db(A)

```
SQL> select pspadm.seq_test.nextval from dual;
```

NEXTVAL
1

b) Wait for 10 secs. Run it on Hash db (B)

```
SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER = 'PSPADM' and SEQUENCE_NAME = 'SEQ_TEST';
```

SEQUENCE_OWNER	SEQUENCE_NAME	LAST_NUMBER
PSPADM	SEQ_TEST	22

c) Wait for 10 secs. Run it on Range db (C)

```
SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER = 'PSPADM' and SEQUENCE_NAME = 'SEQ_TEST';
```

SEQUENCE_OWNER	SEQUENCE_NAME	LAST_NUMBER
PSPADM	SEQ_TEST	43

Prod

1. On Range (current Production)

```
SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER = 'PSPADM' and SEQUENCE_NAME = 'TEST';
```

SEQUENCE_OWNER	SEQUENCE_NAME	LAST_NUMBER
PSPADM	TEST	777

2. Test sequence

```
SQL> select pspadm.test.currval from dual
```

CURRVAL
778

b) Wait for 10 secs. Run it on Hash db (B)

```
SQL> select pspadm.test.nextval from dual;
```

NEXTVAL
798

c) Wait for 10 secs. Run it on Range db (C)

```
SQL> select pspadm.test.nextval from dual;
```

NEXTVAL
820